

⌈ Preliminary.

Lemma. Let $T: V \rightarrow V$ be a linear map on finite-dimensional space V , and let $W \subseteq V$ be a T -invariant subspace.

Suppose $v_1, \dots, v_k$ are eigenvectors of T corresponding to distinct eigenvalues.

If $v_1 + v_2 + \dots + v_k \in W$, then $v_i \in W$ for all $1 \leq i \leq k$.

Proof. If $k=1$, trivial, suppose that for any integer smaller than k , the statement holds.

Assume $v_1, \dots, v_k$ are eigenvectors of T corresponding to distinct eigenvalues λ_i .

If $v_1 + \dots + v_k \in W$, then

$$T(v_1 + \dots + v_k) = \lambda_1 v_1 + \dots + \lambda_k v_k \in W \quad \text{and} \quad \lambda_1(v_1 + \dots + v_k) \in W$$

$\uparrow$ W is T -invariant $\uparrow$ Closed under scalar multiplication.

Subtracting two vectors:

$$\sum_{i=1}^k (\lambda_i - \lambda_k) \cdot v_i = \sum_{i=1}^{k-1} (\lambda_i - \lambda_k) \cdot v_i \in W$$

$i \neq k \Rightarrow \text{not zero}$.

Therefore, $v_i \in W$ for all $1 \leq i \leq k-1$, by the hypothesis. Moreover,

$$v_k = v_1 + \dots + v_k - (v_1 + \dots + v_{k-1}) \in W.$$

Thm. If $T: V \rightarrow V$ is diagonalizable, then for any T -invariant $W \subseteq V$,

$$T|_{W}: W \rightarrow W, x \mapsto T(x)$$

is also diagonalizable.

Proof. Let β be a basis for V s.t. $[T]_{\beta}$ is diagonal.

Let β' be a basis for W . Then, every vector of β' is represented by linear combination of β . That is, if $\beta = \{v_1, \dots, v_n\}$ and $\beta' = \{w_1, \dots, w_k\}$, then for all $1 \leq j \leq k$

$$w_j = a_{1,j} v_1 + \dots + a_{n,j} v_n \quad (a_{i,j} \in F).$$

Consider a n by k matrix $F^k \rightarrow F^n$

$$\begin{pmatrix} [w_1]_{\beta} & \dots & [w_k]_{\beta} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & & & \\ \vdots & & \ddots & \\ a_{n1} & - & - & a_{nk} \end{pmatrix}$$

has rank as k .

Simultaneously Diagonalizable.

In this section, V is finite-dimensional vector space over F , and $T_1, T_2, \dots: V \rightarrow V$ are linear operator.

Def. Linear operators T_1 and T_2 are called simultaneously diagonalizable if:

there exists a basis β for V s.t. both $[T_1]_\beta$ and $[T_2]_\beta$ are diagonal.

Lemma. If T_1, T_2 are simultaneously diagonalizable by a basis β , then for any basis α , $[T_1]_\alpha$ and $[T_2]_\alpha$ are simul. diagonalizable.

Proof. Clearly, $[T_i]_\beta = [T_i]_\alpha^\beta [T_i]_\alpha [T_i]_\beta^\alpha$ thus obtain the result.

Lemma. If T is diagonalizable, then T^m and T are simultaneously diagonalizable, for any $m \geq 1$.

Proof. Let β be a basis for V satisfying $[T]_\beta$ is diagonal. Then

$$[T^m]_\beta = [T]_\beta^m \text{ diagonal.}$$

Thm. If T_1, T_2 are simultaneously diagonalizable by a basis β , then T_1 and T_2 commute. That is, $T_1 \circ T_2 = T_2 \circ T_1$.

Proof. $[T_1 \circ T_2]_\beta = [T_1]_\beta \cdot [T_2]_\beta = [T_2]_\beta \cdot [T_1]_\beta = [T_2 \circ T_1]_\beta$.

Converse

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \cdot \begin{pmatrix} \mu_1 & & 0 \\ & \ddots & \\ 0 & & \mu_n \end{pmatrix} = \begin{pmatrix} \lambda_1 \mu_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \mu_n \end{pmatrix}$$

Thm. If T_1 and T_2 are diagonalizable and commute, then T_1 and T_2 are simultaneously diagonalizable.

Thm. Let $\mathcal{C}$ be a collection of diagonalizable linear operators on V .

there exists a basis β for V s.t. $[T]_\beta$ is diagonal for all $T \in \mathcal{C}$

if and only if

$\mathcal{C}$ commute under composition.



